MAPS BETWEEN MODULES OVER A FLAT COVERING THAT ARE I-POWER TORSION IN KERNEL AND COKERNEL COME FROM MAPS OVER THE BASE RING

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author uses the symbol varphi both for the ambient flat map R -> S and for the given S-module map in the statement and proof; that reuse is reproduced unchanged. The seven references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The gluing category Glue(R -> S, f_1, ..., f_t) and the equivalence of categories with R-modules. invoked at proof lines 24904-24910 and 24920-24921. Is the formalism that converts the descended object into an R-module and the morphism into a map over R.
- I-power torsion modules and their vanishing after localizing at the generators of I. the hypothesis on kernel and cokernel at statement lines 24887-24888 and used at proof line 24903. Makes each localized map an isomorphism, which is what lets the gluing data be assembled.
- The neighbourhood equivalence identifying the kernel and cokernel of a descended map. invoked at proof lines 24951-24952. Gives the final statement comparing kernel and cokernel of varphi and psi.

Source reference. Original lemma 24301: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24301–24301. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original proposition 24813: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24813–24813. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 24641: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24641–24641. Same-chapter navigation pointer;

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the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $\varphi : R \to S$ be a flat ring map and let $I \subset R$ be a finitely generated ideal such that $R/I \to S/IS$ is an isomorphism.*

(1) *Given an R-module N, an S-module M' and an S-module map $\varphi : M' \to N \otimes_R S$ whose kernel and cokernel are I-power torsion, there exists an R-module map $\psi : M \to N$ and an isomorphism $M \otimes_R S = M'$ compatible with φ and ψ.*
(2) *Given an R-module M, an S-module N' and an S-module map $\varphi : M \otimes_R S \to N'$ whose kernel and cokernel are I-power torsion, there exists an R-module map $\psi : M \to N$ and an isomorphism $N \otimes_R S = N'$ compatible with φ and ψ.*

In both cases we have $\mathrm{Ker}(\varphi) \cong \mathrm{Ker}(\psi)$ *and* $\mathrm{Coker}(\varphi) \cong \mathrm{Coker}(\psi)$.

Proof. Proof of (1). Say $I = (f_1, \ldots, f_t)$. It is clear that the localization φ_{f_i} is an isomorphism. Thus we see that $(M', N_{f_i}, \varphi_{f_i}, can_{ij})$ is an object of $\mathrm{Glue}(R \to S, f_1, \ldots, f_t)$, see Remark 24641. By Proposition 24813 we conclude that there exists an R-module M such that $M' = M \otimes_R S$ and $N_{f_i} = M_{f_i}$ compatibly with the isomorphisms φ_{f_i} and can_{ij}. There is a morphism

$$(M \otimes_R S, M_{f_i}, can_i, can_{ij}) = (M', N_{f_i}, \varphi_{f_i}, can_{ij}) \to (N \otimes_R S, N_{f_i}, can_i, can_{ij})$$

of $\mathrm{Glue}(R \to S, f_1, \ldots, f_t)$ which uses φ in the first component. This corresponds to an R-module map $\psi : M \to N$ (by the equivalence of categories of Proposition 24813). The composition of the base change of $M \to N$ with the isomorphism $M' \cong M \otimes_R S$ is φ, in other words $M \to N$ is compatible with φ.

Proof of (2). This is just the dual of the argument above. Namely, the localization φ_{f_i} is an isomorphism. Thus we see that $(N', M_{f_i}, \varphi_{f_i}^{-1}, can_{ij})$ is an object of $\mathrm{Glue}(R \to S, f_1, \ldots, f_t)$, see Remark 24641. By Proposition 24813 we conclude that there exists an R-module N such that $N' = N \otimes_R S$ and $N_{f_i} = M_{f_i}$ compatibly with the isomorphisms $\varphi_{f_i}^{-1}$ and can_{ij}. There is a morphism

$$(M \otimes_R S, M_{f_i}, can_i, can_{ij}) \to (N', M_{f_i}, \varphi_{f_i}, can_{ij}) = (N \otimes_R S, N_{f_i}, can_i, can_{ij})$$

of $\mathrm{Glue}(R \to S, f_1, \ldots, f_t)$ which uses φ in the first component. This corresponds to an R-module map $\psi : M \to N$ (by the equivalence of categories of Proposition 24813). The composition of the base change of $M \to N$ with the isomorphism $N' \cong N \otimes_R S$ is φ, in other words $M \to N$ is compatible with φ.

The final statement follows for example from Lemma 24301. □